

SYNOPSIS



At 0449 on 29 October 2008, the UK registered general cargo vessel *Scot Isles* was in collision with the Egyptian bulk carrier *Wadi Halfa* in the Dover Strait.

The watchkeeping officer on *Scot Isles*, which was on passage from Rochester to Antwerp and crossing the NE traffic lane of the Dover Strait Traffic Separation Scheme, did not detect *Wadi Halfa* before the collision.

The watchkeeping officer on *Wadi Halfa*, which was on a NE course in the NE traffic lane, saw *Scot Isles* when she was very close but, despite taking evasive action, could not prevent the collision.

Scot Isles suffered extensive shell plate damage on her starboard side which resulted in the loss of 60 tonnes of marine gas oil into the sea. *Wadi Halfa* was damaged, less severely, on her port side and was able to resume her passage to Bremen.

The vessels were within French coastguard jurisdiction when the accident was reported and, once the extent of damage and pollution was realised, the French authorities directed *Scot Isles* to proceed to Dunkirk to facilitate repairs and an investigation into the accident.

As a result of a complacent attitude to bridge watchkeeping on both vessels, safety barriers, which would have warned the bridge watchkeeping officers of the risk of a collision, were not in place. No lookout was present on either bridge at the time of the collision, and the vessels' radars and other bridge equipment were not used effectively.

A Safety Flyer has been published which identifies the key safety lessons from this investigation. Recommendations have been made to the International Chamber of Shipping, the UK Chamber of Shipping, The Nautical Institute and the International Federation of Shipmasters' Associations designed to promulgate the contents of the Safety Flyer to those organisations' membership to highlight to ship operators and seafarers the importance of effective bridge teams and the maintenance of proper navigational lookouts.

1.2 BACKGROUND

Scot Isles, a UK registered, single hold general cargo vessel, operated in NW European waters and was engaged primarily to carry timber and timber products from Scandinavia. On the morning of the collision, she was crossing the Dover Strait Traffic Separation Scheme (TSS) from the Thames estuary to Antwerp.

Wadi Halfa, an Egyptian registered, bulk/general cargo vessel, traded worldwide. She was in ballast, on passage from Egypt to Germany, in the NE traffic lane when the collision occurred.

1.3 NARRATIVE

1.3.1 *Scot Isles*

Scot Isles arrived in Rochester at 0100 on 28 October 2008 to discharge her cargo of timber. Cargo discharge began at 0800 and was completed by 1530. The chief and second officers shared the cargo watches in port. The chief officer was on duty from 0800 - 1200 when he was relieved by the second officer.

During the day, the vessel's classification society was changed and the surveys and documentation associated with this process occupied much of the master's time from 0900 until the required cargo and related clearance documentation was completed at 1900. The vessel then awaited the tide before sailing.

The vessel sailed at midnight, in ballast, bound for Antwerp. The master held a pilotage exemption certificate (PEC) for the Medway and took the conn of the vessel for departure. On the bridge with him were the second officer, who was the 0000 - 0400 watchkeeping officer and a lookout, who kept the 0000 - 0600 watch.

At 0150 on 29 October, the vessel left the Medway pilotage area and entered the Thames estuary. The master then handed over the conn to the second officer and left the bridge. The bridge watch alarm was not activated.

Scot Isles cleared the Thames estuary and at 0324 the second officer set a course of 105°(T) as the vessel approached the northern Dover Strait (**Figure 1**).

At 0400, the vessel was 4 miles from the SW traffic lane of the Dover Strait TSS, when the chief officer took over as the bridge watchkeeping officer. He adjusted the set up of the starboard radar, which was equipped with an Automatic Radar Plotting Aid (ARPA), to the 6 mile range scale, ship's head up, relative motion. He offset the centre of the radar display to provide a range of about 9 miles ahead. The port radar was not in use.

Although there were several targets on the radar screen, none of them was acquired on ARPA to assess whether a risk of collision existed.

The Automatic Identification System (AIS), located beside the starboard radar and which provided the names and relative positions of the six nearest vessels to *Scot Isles*, was not being monitored.

At 0410, *Scot Isles* altered course to 109°(T), and at 0421 she began to cross the SW traffic lane of the TSS. The vessel's position was not plotted on the chart after 0400, so this reconstruction is based on radar replay information obtained from the Channel Navigation Information Service (CNIS) at Dover following the accident (**Figure 2**).

AIS data courtesy of the MCA

Figure 2



Radar replay screen shot from CNIS, 0449 29 October 2008

At about 0435 the lookout reported to the chief officer, who was sitting in the wheelhouse chair (**Figure 3**), the red navigation light of a vessel at about 3 points (34°) to starboard of *Scot Isles*. The lookout then advised the chief officer, who made no comment, that he was leaving the bridge to undertake his safety rounds. The chief officer had not registered the lookout's report of the other vessel.

At 0443 *Scot Isles* began to cross the NE traffic lane (**Figure 1**).

The chief officer became aware of the accommodation lights of a vessel very close on *Scot Isles*' starboard quarter. He moved towards the wheel, intending to put the helm to port, when at 0449, the collision occurred.



Scot Isles - interior of wheelhouse

1.3.2 Wadi Halfa

Wadi Halfa sailed from Alexandria, Egypt, on 17 October. She was in ballast with a draught of 5.29m bound for Bremen, Germany.

At about 0215 on 29 October, the master arrived on the bridge, as previously arranged, to be present as the vessel passed Cap Griz Nez and entered the Dover Strait, in the NE traffic lane (**Figure 1**). At 0345, as the vessel entered the Sandettie deep water route, the master left the bridge.

The chief officer relieved the second officer as the bridge watchkeeping officer at 0400. Also on the bridge for the 0400 - 0800 watch were a cadet and an able seaman (AB). The cadet was being trained in navigation and the AB was the designated lookout for the watch.

At 0430 the vessel altered course to 041°(T) and, at about 0435, the chief officer allowed the lookout to leave the bridge for a toilet break. The cadet had remained in the chart room area, which was curtained off from the wheelhouse (**Figure 4**) to prevent light pollution, from 0400. He is reported to have been recording the vessel's position every 10 minutes using the vessel's Global Positioning System (GPS) display.

1.8 ENVIRONMENTAL CONDITIONS

Wind:	Northerly, Force 3 (Beaufort scale)
Sea:	Slight to Moderate
Visibility:	Good
Weather:	Occasional rain showers
Tide:	Slack water

1.9 CHANNEL NAVIGATION INFORMATION SERVICE (CNIS)

Dover Strait is one of the busiest shipping lanes in the world. Vessels are tracked and recorded by both radar and AIS. Shipping movements in the area are monitored by the English and French coastguard from radar stations located at Dover and Cap Gris Nez respectively.

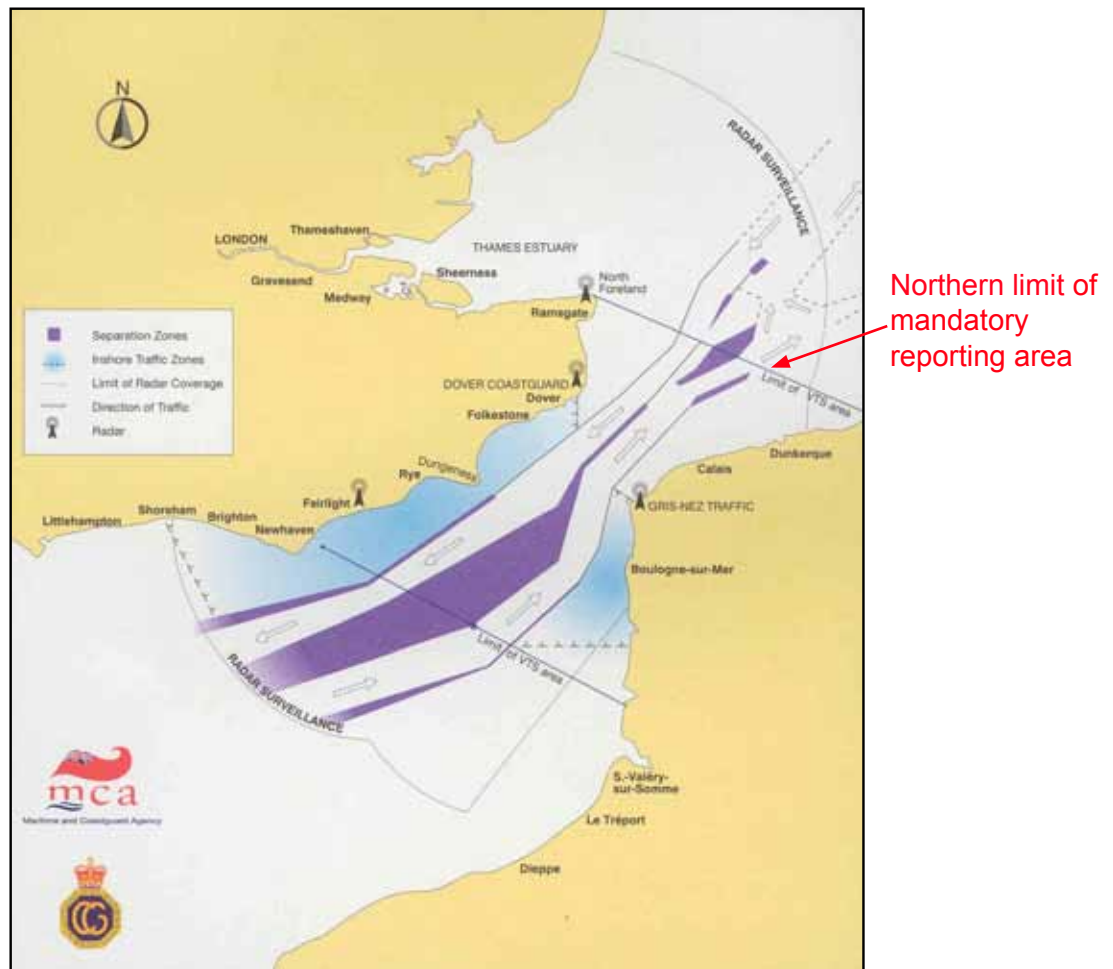
Dover and Cap Gris Nez provide a coastal vessel traffic information service. Each station broadcasts information on VHF radio about weather and navigational hazards as part of the joint CNIS. Their broadcasts also include information on deep draught vessels, vessels under tow and on vessels engaged in special operations. A broadcast is also transmitted for any vessel that appears to be in contravention of the COLREGS to warn other vessels that a potentially hazardous situation exists in a particular part of the TSS.

A mandatory reporting scheme for vessels over 300GT using the TSS was introduced in 1999. SW bound vessels are required to report to Dover coastguard and NE bound vessels report to Cap Gris Nez.

To assist mariners transiting Dover Strait, the MCA published Marine Guidance Note (MGN) 364 (M+F), which contains specific reference to the Sandettie deep water route (**Annex B**).

Wadi Halfa reported to Cap Gris Nez when she entered the NE bound traffic lane, while *Scot Isles* was not required to make a report as her passage was conducted outside of the mandatory reporting area (**Figure 9**).

The tracks of both vessels were recorded on radar by CNIS at Dover and the radar plot for 0449, 29 October (**Figure 2**) shows their historic tracks at the time of the collision.



Dover Strait TSS - mandatory reporting area

1.10 KEEPING A SAFE NAVIGATIONAL WATCH

A number of publications contain relevant advice for masters and watchkeeping officers on the essentials of keeping a safe navigational watch. Among these are:

- MGN 315 (M) – Keeping a safe navigational watch on merchant vessels
- MGN 137 (M+F) – Lookout during periods of darkness and reduced visibility
- STCW Chapter 8 – Standards regarding watchkeeping.

All of the above publications emphasise the need to maintain a proper lookout at all times. MGN 315 (M) (**Annex F**) provides detailed guidance on both the requirements for posting a lookout and the relationship between the bridge watchkeeping officer and the lookout.

1.10.1 COLREGS

Rule 5 defines the duty to maintain a proper lookout:

Every vessel shall at all times maintain a proper look-out by sight and hearing as well as by all available means appropriate in the prevailing circumstances and conditions so as to make a full appraisal of the situation and of the risk of collision.

Rule 15 defines the action to be taken in a crossing situation:

When two power-driven vessels are crossing so as to involve risk of collision, the vessel which has the other on her own starboard side shall keep out of the way and shall, if the circumstances of the case admit, avoid crossing ahead of the other vessel.

The other rules considered relevant to this accident are included in full at **Annex G**.

1.11 PREVIOUS INCIDENTS

In 2004 the MAIB published a Bridge Watchkeeping Safety Study¹ which reviewed 1,647 collisions, groundings, contacts and near collisions involving merchant vessels over 500GT reported to the MAIB between 1994 and 2003.

In relation to collisions, the study found that:

- 65% of vessels involved in collisions were not keeping a proper lookout
- 33% of all accidents that occurred at night involved a sole watchkeeper on the bridge
- on 19% of the vessels involved in the collisions, the bridge watchkeeping officers were completely unaware of the other vessel until or, in some cases, after the collision.

In November 2004 *Scot Explorer*, which was also managed by Intrada Ships Management Limited, collided with the fishing vessel *Dorthe Dalsøe*.

The MAIB carried out a full investigation of this accident (report 10/2005 refers)².

The report concluded that, while the fishing vessel was the give-way vessel and should have kept clear, *Scot Explorer* did not maintain a proper lookout and should have made better use of her radar, including ARPA, to facilitate an earlier determination that a risk of collision existed. At the time of the collision the designated lookout on *Scot Explorer*, who was also the ship's cook, was in the galley preparing a meal.

¹ MAIB Safety Study 1/2004,
http://www.maib.gov.uk/publications/safety_studies/bridge_watchkeeping_safety_study.cfm

² MAIB Investigation Report,
http://www.maib.gov.uk/publications/investigation_reports/2005/scot_explorer.cfm

A recommendation was made to the International Chamber of Shipping to highlight to ship owners that the accident could have been avoided if a dedicated lookout had been on the bridge during hours of darkness.

In September 2008, *Scot Venture*, managed by Intrada Ships Management Limited, collided with the fishing vessel *Golden Promise*. The MAIB carried out a Preliminary Examination into the accident³ and Intrada issued a safety circular (**Annex D**) reminding bridge officers of the principles of good watchkeeping practice.

1.12 BRIDGE RESOURCE MANAGEMENT

The need to maintain an effective bridge team at all times is one of the key themes of the Bridge Procedures Guide, issued by the International Chamber of Shipping⁴. The guide states that, inter alia:

At all times, ships need to be navigated safely in compliance with the COLREGS and also to ensure that protection of the marine environment is not compromised.

An effective bridge organisation should manage efficiently all the resources that are available to the bridge and promote good communication and teamwork.

The need to maintain a proper look-out should determine the basic composition of the navigational watch. There are, however, a number of circumstances and conditions that could influence at any time the actual watchkeeping arrangements and bridge manning levels.

Effective bridge resource and team management should eliminate the risk that an error on the part of one person could result in a dangerous situation.

³ MAIB Preliminary Examination,
http://www.maib.gov.uk/publications/completed_preliminary_examinations/completed_preliminary_examinations_2008/scot_venture.cfm

⁴ ICS Bridge Procedures Guide, Fourth Edition, 2007

SECTION 2 - ANALYSIS

2.1 AIM

The purpose of the analysis is to determine the contributory causes and circumstances of the accident as a basis for making recommendations to prevent similar accidents occurring in the future.

2.2 SCOT ISLES

2.2.1 Master

At 0150 on the morning of the collision, when the vessel had cleared the Medway pilotage area, the master left the second officer as the sole watchkeeping officer for the passage in the Thames estuary. He considered that Thames VTS and CNIS would alert the vessel to any potentially hazardous situations and would provide advice on collision avoidance situations.

It is not the role of a VTS to relieve vessels of their duties under the COLREGS to maintain a proper lookout and to take proper action to prevent collision. VTS provides navigational safety information to vessels in its area, but the service should not be regarded as a substitute for maintaining a safe navigational watch at all times.

The master did not leave night orders, and did not normally do so. It was his assumption that, because the bridge watchkeeping officers all held certificates of competency, they did not require any further guidance.

Night orders were required to be completed by the vessel's SMS and are an effective way of alerting watchkeepers to specific hazards at any stage of a passage. They provide an additional safety barrier to prevent unsafe practices and act as a stimulus to bridge watchkeeping officers by raising their awareness of the risks likely to be encountered and by providing guidance on how to mitigate those risks.

In this case night orders could have been used to identify key areas of concern relating to the passage across the traffic lanes of the Dover Strait TSS. In this respect the master's requirements with reference to: maintaining a proper lookout, desired interval for position fixing on the chart, use of ARPA for radar plotting, safe passing distances from other vessels and navigational marks, as well as the effective use of AIS could have been listed.

Evidence also indicates that it was the master's custom not to post a lookout during his bridge watches and that he did not require the bridge watch alarm to be used.

The manager had issued a fleet directive on keeping a safe navigational watch and a safety circular to bridge watchkeepers referring to the Bridge Procedures Guide. The MCA provides guidance in MGN 315(N) on maintaining a safe navigational watch.

Despite his knowledge and experience, the specific requirements contained in the manager's SMS and the national and international recommendations with respect to bridge watchkeeping, the master had become complacent. This factor, which had not been identified by the manager, set a low benchmark for others on board *Scot Isles* to follow.

2.2.2 Chief officer's watchkeeping

The chief officer took over as the bridge watchkeeping officer at 0400 on a dark night in relatively benign weather conditions. The vessel was clear of the Thames estuary and there were only a few other vessels in the vicinity.

As the vessel proceeded towards the SW traffic lane, in open water, the chief officer was faced with no immediate risks to the safe passage of the vessel. The Filipino lookout was on the bridge, but it was not customary for the watchkeeping officers to talk to the lookouts. When the lookout reported a red light to starboard, and then advised that he was leaving the bridge to undertake safety rounds, the chief officer made no comment. As this was not abnormal, the lookout was unaware that the chief officer had not registered his report.

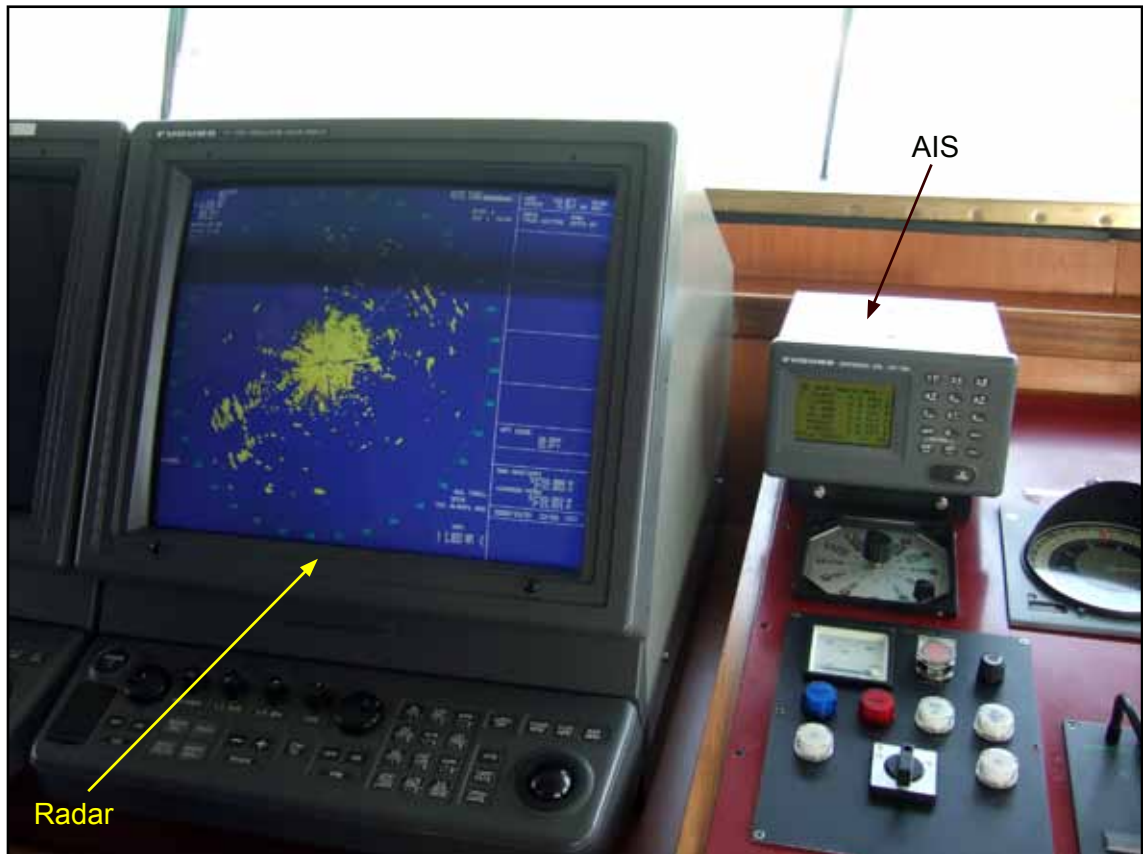
The absence of a lookout demonstrated a complacent attitude to watchkeeping when the vessel was crossing the traffic lanes of a TSS.

Safe crossing of a traffic lane in a TSS requires the bridge team to be alert and to maintain situational awareness. This is achieved by effective bridge resource management, which was lacking in this case. The lookout should not have been allowed to leave the bridge when he did.

The chief officer was sitting in front of a radar fitted with ARPA, but this facility was not used to acquire and plot any of the radar targets, which could have enabled an early assessment of risk of collision to be made. The information displayed on the AIS receiver (**Figure 10**), located beside the starboard radar, giving the names and relative positions of the closest six vessels, was also not used.

The chief officer failed to maintain a proper lookout by sight or use of the navigational equipment available to him. This was contrary to Rule 5 of the COLREGS.

Figure 10



Scot Isles - wheelhouse, location of AIS receiver

The vessel's position was not plotted on the chart after 0400. The GPS display was monitored to ensure the vessel remained within the tram lines displayed on screen. This was poor navigational practice, as reference to a chart would have helped the chief officer to maintain situational awareness. Regular use of the chart provides the watchkeeper with a stimulus from the essential information it contains to assist the conduct of a safe passage.

Analysis of the damage caused to both vessels indicates that the chief officer probably did not see *Wadi Halfa* before *Scot Isles* was struck amidships.

He first became aware of the other vessel when he looked out of the starboard aft facing wheelhouse window and saw the accommodation lights of *Wadi Halfa* just before the second impact, when she struck the starboard bridgework of *Scot Isles*.

There is no evidence to conclude that the chief officer was unduly fatigued or that he fell asleep during the period leading up to the collision. However, it is likely that his apparent lethargic approach to his watchkeeping responsibilities was due to a lack of stimuli because of the following:

SECTION 3 - CONCLUSIONS

3.1 SAFETY ISSUES DIRECTLY CONTRIBUTING TO THE ACCIDENT WHICH HAVE RESULTED IN RECOMMENDATIONS

1. The masters of both vessels were complacent, a factor that had not been identified by the respective ship managers and which did not engender best practice in either bridge team. [2.2.1, 2.3.1]
2. It is likely that *Scot Isles*' chief officer's apparent lethargic approach to his watchkeeping responsibilities was due to a lack of stimuli:
 - There was no routine interaction between him and his dedicated lookout, who, in any case, had been allowed to leave the bridge over 10 minutes before the collision.
 - There were no night orders left by the master to alert the watchkeeper to the risks of the passage across the TSS and the requirement to be extra vigilant.
 - ARPA was not used to acquire and plot targets
 - Positions were not plotted on the chart
 - The AIS was not monitored
 - The bridge watch alarm was not activated
 - He remained seated throughout
 - He had become complacent in his approach to watchkeeping [2.2.2]
3. The chief officer of *Wadi Halfa* demonstrated a complacent attitude to his bridge watchkeeping responsibilities and could have made better use of available resources. [2.3.2]
4. When the lookouts were allowed to leave their respective bridges, neither chief officer realised that a significant safety barrier had been removed. On both vessels, available navigational equipment was not used effectively and provided an inadequate substitute for maintaining a proper lookout by sight. [2.4.1]
5. Safety management system non conformities on each vessel with respect to the formation of effective bridge teams and the use of lookouts had not been identified during internal company audits. [2.8]

3.2 SAFETY ISSUES IDENTIFIED DURING THE INVESTIGATION WHICH HAVE NOT RESULTED IN RECOMMENDATIONS BUT HAVE BEEN ADDRESSED

1. An instruction to sound the General Alarm was not included in the SMS post collision checklist on *Scot Isles*. [2.5.1]
2. *Scot Isles* did not advise the nearest Coastal State of the accident. [2.5.1]
3. The post collision checks carried out on *Scot Isles* were not sufficient to identify the major damage to the hull. [2.5.1]